

EXTENSION 2

Complex Numbers (Ext2), N1 Introduction to Complex Numbers (Ext2)

Exponential Form

Teacher: Nardene Dodd

Exam Equivalent Time: 24 minutes (based on allocation of 1.5 minutes per mark)



Questions

1. Complex Numbers, EXT2 N1 EQ-Bank 5 MC

In which quadrant of the complex plane is the complex number $4e^{\frac{11\pi}{3}}$ found?

- A. I
- B. II
- C. III
- D. IV

2. Complex Numbers, EXT2 N1 SM-Bank 1 MC

Which of the following is the complex number $-\sqrt{3} + 3i$?

- A. $2\sqrt{3}e^{-\frac{i\pi}{3}}$
- B. $2\sqrt{3}e^{\frac{i2\pi}{3}}$
- C. $12e^{-\frac{i\pi}{3}}$
- D. $-12e^{\frac{i2\pi}{3}}$

3. Complex Numbers, EXT2 N1 EQ-Bank 3 MC

Which of the following is the complex number $-3 - \sqrt{3}i$?

- A. $12e^{-\frac{i5\pi}{6}}$
- B. $12e^{\frac{i\pi}{6}}$
- C. $2\sqrt{3}e^{-\frac{i5\pi}{6}}$
- D. $2\sqrt{3}e^{\frac{i\pi}{6}}$

4. Complex Numbers, EXT2 N1 2021 HSC 11a

The complex numbers $z = 2e^{i\frac{\pi}{2}}$ and $w = 6e^{i\frac{\pi}{3}}$ are given.

Find the value of zw , giving the answer in the form $re^{i\theta}$. (2 marks)

5. Complex Numbers, EXT2 N1 2022 HSC 11c

- i. Write the complex number $-\sqrt{3} + i$ in exponential form. (2 marks)
- ii. Hence, find the exact value of $(-\sqrt{3} + i)^{10}$ giving your answer in the form $x + iy$. (2 marks)

6. Complex Numbers, EXT2 N1 EQ-Bank 2

Express the complex number $z = -2\sqrt{2} - 2\sqrt{6}i$ in the exponential form. (2 marks)

7. Complex Numbers, EXT2 N1 EQ-Bank 9

Calculate the value of $\frac{e^{\frac{i\pi}{3}} - e^{-\frac{i\pi}{3}}}{2i}$. (2 marks)

8. Complex Numbers, EXT2 N1 EQ-Bank 6

Given $z = \frac{-1 - i\sqrt{3}}{1 + i}$, calculate z^2 in exponential form. (3 marks)

Worked Solutions

1. Complex Numbers, EXT2 N1 EQ-Bank 5 MC

$$z = 4e^{\frac{i16}{3}}$$

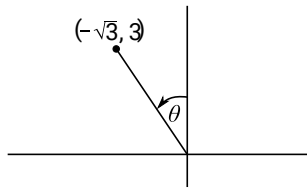
$$\arg(z) = \frac{16}{3} \approx 5.3$$

$$\frac{3\pi}{2} < 5.3 < 2\pi$$

$\therefore$ Quadrant IV

$\Rightarrow D$

2. Complex Numbers, EXT2 N1 SM-Bank 1 MC



$$|z| = \sqrt{(\sqrt{3})^2 + 3^2} = 2\sqrt{3}$$

$$\tan \theta = \frac{\sqrt{3}}{3} = \frac{1}{\sqrt{3}}$$

$$\theta = \frac{\pi}{6}$$

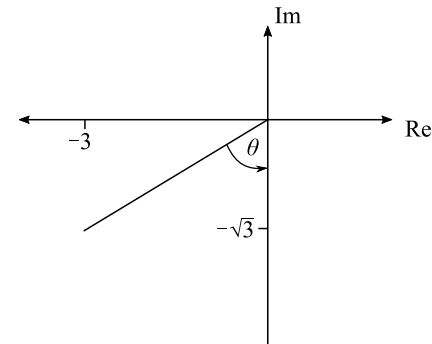
$$\arg(z) = \frac{\pi}{2} + \frac{\pi}{6} = \frac{2\pi}{3}$$

$$\begin{aligned} \therefore z &= 2\sqrt{3} \left(\cos\left(\frac{2\pi}{3}\right) + i\sin\left(\frac{2\pi}{3}\right) \right) \\ &= 2\sqrt{3}e^{\frac{i2\pi}{3}} \end{aligned}$$

$\Rightarrow B$

Worked Solutions

3. Complex Numbers, EXT2 N1 EQ-Bank 3 MC



$$\begin{aligned} |z| &= \sqrt{3^2 + (\sqrt{3})^2} \\ &= \sqrt{12} \\ &= 2\sqrt{3} \end{aligned}$$

$$\tan \theta = \frac{3}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \sqrt{3}$$

$$\theta = \frac{\pi}{3}$$

$$\arg(z) = -\left(\frac{\pi}{3} + \frac{\pi}{2}\right) = \frac{5\pi}{6}$$

In exponential form:

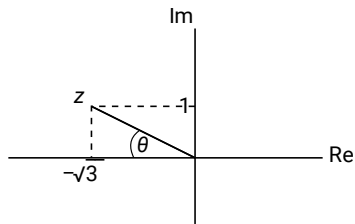
$$\begin{aligned} z &= 2\sqrt{3}e^{-\frac{i5\pi}{6}} \\ &\Rightarrow C \end{aligned}$$

4. Complex Numbers, EXT2 N1 2021 HSC 11a

$$\begin{aligned} zw &= 2e^{i\frac{\pi}{2}} \cdot 6e^{i\frac{\pi}{6}} \\ &= 12e^{i\frac{\pi}{2} + i\frac{\pi}{6}} \\ &= 12e^{i\frac{2\pi}{3}} \end{aligned}$$

5. Complex Numbers, EXT2 N1 2022 HSC 11c

i.



$$\text{Let } z = -\sqrt{3} + i$$

$$|z| = \sqrt{(-\sqrt{3})^2 + 1^2} = 2$$

Find $\arg(z)$:

$$\tan \theta = \frac{1}{\sqrt{3}} \Rightarrow \theta = \frac{\pi}{6}$$

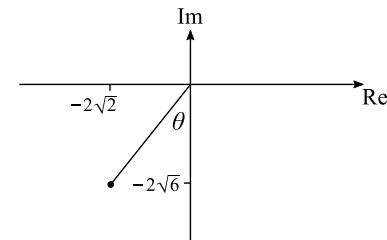
$$\Rightarrow \arg(z) = \frac{5\pi}{6}$$

$$\begin{aligned} \therefore z &= 2 \left(\frac{\cos(5\pi/6)}{6} + \frac{\sin(5\pi/6)}{6} i \right) \\ &= 2e^{\frac{5\pi}{6}i} \end{aligned}$$

$$\begin{aligned} \text{ii. } (-\sqrt{3} + i)^{10} &= \left(2e^{\frac{5\pi}{6}i} \right)^{10} \\ &= 2^{10} e^{\frac{50\pi}{6}i} \\ &= 1024 e^{\frac{\pi}{3}i} \\ &= 1024 \left(\cos\left(\frac{\pi}{3}\right) + \sin\left(\frac{\pi}{3}\right) i \right) \\ &= 1024 \left(\frac{1}{2} + \frac{\sqrt{3}}{2} i \right) \\ &= 512 + 512\sqrt{3}i \end{aligned}$$

6. Complex Numbers, EXT2 N1 EQ-Bank 2

$$\begin{aligned} |z| &= \sqrt{(2\sqrt{2})^2 + (2\sqrt{6})^2} \\ &= \sqrt{8 + 24} \\ &= 4\sqrt{2} \end{aligned}$$



$$\begin{aligned} \tan \theta &= \frac{2\sqrt{2}}{2\sqrt{6}} = \frac{1}{\sqrt{3}} \\ \theta &= \frac{\pi}{6} \end{aligned}$$

$$\text{Arg}(z) = -\left(\frac{\pi}{2} + \frac{\pi}{6}\right) = -\frac{2\pi}{3}$$

$$\begin{aligned} \therefore z &= 4\sqrt{2} \text{cis}\left(-\frac{2\pi}{3}\right) \\ &= 4\sqrt{2} e^{-i\frac{2\pi}{3}} \end{aligned}$$

7. Complex Numbers, EXT2 N1 EQ-Bank 9

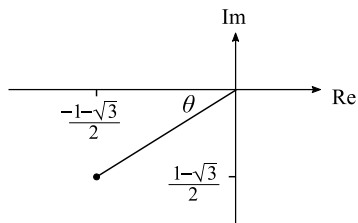
$$\begin{aligned} \frac{e^{\frac{i\pi}{3}} - e^{-\frac{i\pi}{3}}}{2i} &= \frac{\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} - (\cos(-\frac{\pi}{3}) + i \sin(-\frac{\pi}{3}))}{2i} \\ &= \frac{\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} - (\cos \frac{\pi}{3} - i \sin \frac{\pi}{3})}{2i} \\ &= \frac{2i \sin \frac{\pi}{3}}{2i} \\ &= \sin \frac{\pi}{3} \\ &= \frac{\sqrt{3}}{2} \end{aligned}$$

8. Complex Numbers, EXT2 N1 EQ-Bank 6

$$\begin{aligned} z &= \frac{-1-i\sqrt{3}}{1+i} \times \frac{1-i}{1-i} \\ &= \frac{(-1-i\sqrt{3})(1-i)}{1-i^2} \\ &= \frac{-1+i-i\sqrt{3}+i^2\sqrt{3}}{2} \\ &= \frac{-1-\sqrt{3}}{2} + i \left(\frac{1-\sqrt{3}}{2} \right) \end{aligned}$$

$$\begin{aligned} |z| &= \sqrt{\left(\frac{-1-\sqrt{3}}{2} \right)^2 + \left(\frac{1-\sqrt{3}}{2} \right)^2} \\ &= \sqrt{\frac{1+2\sqrt{3}+3+1-2\sqrt{3}+3}{4}} \\ &= \sqrt{2} \end{aligned}$$

Find $\arg(z)$:



$$\tan \theta = \frac{\frac{1-\sqrt{3}}{2}}{\frac{-1-\sqrt{3}}{2}} = \frac{\sqrt{3}-1}{\sqrt{3}+1}$$

$$\theta = 15^\circ = \frac{\pi}{12}$$

$$\therefore \arg(z) = -\frac{11\pi}{12}$$

$$z = \sqrt{2} \text{cis} \left(-\frac{11\pi}{12} \right)$$

$$\begin{aligned} z^2 &= \left(\sqrt{2} \right)^2 \text{cis} \left(-\frac{11\pi}{12} \times 2 + 2\pi \right) \\ &= 2 \text{cis} \left(\frac{\pi}{6} \right) \\ &= 2e^{\frac{i\pi}{6}} \end{aligned}$$